Problem Set 2

Two sets X, Y are said to have the same *cardinality* (cardinal, cardinal number), denoted $|X| = |Y|$, if there exists a bijection $f : X \to Y$. For $n \in \mathbb{N}$, we let $[n] = \{1, 2, \ldots, n\}$. We say that a set X is finite if $|X| = |[n]|$ for some $n \in \mathcal{N}$. Since $|[n]| = |[m]|$ if and only if $n = m$, we have define finite cardinals as natural numbers, i.e., $|[n]| = n$ for all $n \in \mathbb{N}$.

We can define an ordering of cardinal numbers. We say that $|X| \leq |Y|$ if there is an injection $f : X \to Y$. We also define the strict ordering $|X| < |Y|$ to mean that $|X|| \leq Y$ but $|X| \neq |Y|$.

We say that a set X is *countably infinite* if there is a bijection $\mathbb{N} \to X$, i.e., $|\mathbb{N}| = |X|$. A set is called *countable* if it is finite or countably infinite. We say that X is *uncountable* if it is not countable.

Starter

1 (M)**.** Let A be a set. Show that the following are equivalent:

(i) A is countable.

(ii) There exists an injection $A \to \mathbb{N}$.

(iii) $A = \varnothing$ or there exists a surjection $\mathbb{N} \to A$.

2 (E)**.** Show that any subset of a countable set is countable.

3 (E)**.** (a) Prove that $\mathbb{Q}$ is countable.

(b) Prove that $\mathbb{R}$ is uncountable.

(c) Prove that the union of two countable sets is countable.

(d) Prove that $\mathbb{R} \setminus \mathbb{Q}$ is uncountable.

4 (E)**.** Prove or disprove each statement. For a false statement, give a counterexample and justify why it is a counterexample.

(a) Every subset of an uncountable set is uncountable.

(b) If A and B are uncountable, then $A \cap B$ is uncountable.

(c) If A is uncountable, then $A \cup B$ is uncountable for every set B.

Advanced

5 (L-M)**.** Construct a bijection from $(0, 1)$ to $[0, 1]$.

6 (M)**.** Show that countably union of countable sets is countable.

7 (M)**.** A function f with domain $\mathbb{N}$ is called *eventually periodic* if there are natural numbers n_0 and $k \geq 1$ such that $f(n + k) = f(n)$ for all $n \geq n_0$. Let C be the collection of eventually periodic functions $f : \mathbb{N} \to \mathbb{N}$. Prove that C is countable.

8 (H)**.** Prove the Cantor-Schröder–Bernstein Theorem: If X and Y are sets such that $|X| \leq |Y|$ and $|Y| \leq |X|$, then $|X| = |Y|$.